

Hill Choy

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SUMMARY

MSc student at University College London with a BEng in Computer Engineering from HKUST. Experienced in developing and optimizing Computer Vision models for real-time Edge AI applications, including traffic and pedestrian monitoring systems. Focused on the practical implementation of machine learning to solve real-world visual computing challenges.

EDUCATION

University College London (UCL)

MSc Computer Graphics, Vision and Imaging

Expected Graduation: September 2026

○ Highlighted Coursework:

- COMP0137 Machine Vision
- COMP0169 Machine Learning for Visual Computing
- COMP0263 Solving Inverse Problems with Data-Driven Models
- COMP0249 Robot Vision and Navigation

Hong Kong University of Science and Technology (HKUST)

BEng in Computer Engineering (Second Class Honors, Division 1)

Graduation: July 2024

○ Highlighted Coursework:

- COMP4221 Introduction to Natural Language Processing
- COMP4621 Computer and Communication Networks

IELTS Academic

8 (Out of 9), (R:8.5,L:9,W:7,S:7)

RESEARCH EXPERIENCE

Advanced Video Analytics and Edge AI for Smart Carpark Systems

HKUST, Bachelor Degree Final Year Project (Supervisor: Prof. Shueng-Han Gary Chan, CSE, HKUST)

- Developed a smart car park system that utilizes edge AI to provide real-time parking space availability. It is capable of informing drivers of available parking spaces while minimizing the computational, installation, and maintenance costs of the system.
 - Utilized You Only Look Once (YOLO) series models for the detection of cars on Raspberry Pi.
 - Implemented a MobileNetV3-based backbone in YOLOv8 to prune the model size while maintaining performance.

PROFESSIONAL EXPERIENCE

Research Assistant

Prof. Shueng-Han Gary Chan, CSE, HKUST

Hong Kong

08/2024 - 06/2025

- Developed computer vision-based solutions using YOLO series models to meet the various needs of projects.
- Developed web applications using Python and Flask for processing and displaying information.
- Optimized models and application pipelines for computer vision on IoT devices.
- List of Projects:
 - K11 MUSEA Cars and Pedestrian Flow Monitoring System
 - Monitor real-time traffic and pedestrian flow in key areas of the business-mall complex.
 - Provide property management with daily, weekly, monthly, and yearly reports to facilitate management.
 - Responsible for the training of models and development of data capture pipelines, backend, and the web frontend.
 - Privacy-Preserving Elderly Monitoring with a Novel Edge AI Camera
 - Monitor patients/elderly daily using privacy-conserving technology and alert staff if a patient or elderly person falls.
 - Recipient of the **Hong Kong ICT Awards 2024, Smart People (Smart Ageing) Silver Award**.
 - Responsible for data visualization and daily reports generation scripts for patients and the elderly.
 - Smart Traffic Management System (Kung Tong, Hong Kong)
 - Monitored real-time traffic at major junctions in the Kung Tong District, alerting authorities to congestion and accidents.
 - Responsible for developing real-time traffic information capture pipelines for traffic patterns analytics.
 - Link: <https://www.police.gov.hk/>

ADDITIONAL

- **Languages:** Cantonese(Native), Mandarin(Basic), English(Fluent), Japanese(Foundation)